

Prithvi Chakravarthy

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EDUCATION

DePaul University

Master of Science in Computer Science

Chicago, IL

Sept. 2025 – Present

- Coursework: Distributed Systems, Applied Algorithms and Structures, DBMS, Real-Time System Architecture

BITS Pilani, WILP

Postgraduate Certificate in AI/ML

India

Oct. 2023 – Oct. 2024

- Coursework: Deep Learning, Neural Networks, Text Mining, Feature Engineering, Unsupervised Learning

VIT Bhopal

Bachelor of Technology in Computer Science

India

Jun. 2018 – Jun. 2022

- Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Architecture, Soft Computing, Computer Networks, Theory of Computation & Compiler Design

EXPERIENCE

Designare Solutions Pvt Ltd

Full Stack Developer

Mumbai, India

March 2022 – August 2025

- Engineered and deployed a diverse portfolio of full-stack applications, including **Geospatial (GIS)**, **Social Media (Mobile)**, and **Booking** platforms using React, Node.js, Python, and Kotlin.
- Architected scalable REST and **Distributed APIs**, optimizing complex spatial and relational queries to achieve a consistent **25% reduction in data retrieval latency** across enterprise platforms.
- Managed cloud-native infrastructure on **Google Cloud Platform (GCP)**, implementing **Cron-driven automation** for data synchronization and maintaining **99.9% uptime** for high-traffic services.
- Integrated advanced behavioral analytics (Amplitude/Google Analytics) and implemented enterprise-grade security protocols, including **Firebase Authentication** and **Role-Based Access Control (RBAC)**.

PROJECTS

Deepstory (Social Media Platform) | Kotlin/Android, PostgreSQL, GCP, Firebase, Amplitude

- Developed a short-form video app with **Clean Architecture**, creating APIs for streaming, likes, and profiles.
- Engineered backend synchronization using **Cron jobs** for periodic updates of likes, views, and comments.

Space Invaders Game Engine | C#, Azul Engine, GoF Patterns

- Architected a C# engine implementing **14+ GoF design patterns** and **SOLID principles**
- Eliminated runtime heap allocation via an **Object Pool** system and used **Proxy/Flyweight** for sprite rendering.
- Implemented a decoupled collision system using **Visitor and Observer** patterns with double-dispatch.

AI Journal OS | Rust, Tauri, React, TypeScript, SQLite

- Built a privacy-first, local-offline AI journaling app: a Tauri/React frontend paired with a standalone **Rust** background worker in a **dual-topology architecture**.
- Engineered a local **RAG** pipeline with pure Rust vector math, SQLite BLOB storage, and **all-MiniLM-L6-v2** embeddings.
- Designed a pipeline that redacts PII locally via **Hugging Face** models before LLM API calls, with **AES-256 encrypted** zero-knowledge cloud sync.

Stable Diffusion Image Generation Model | Python, PyTorch, Jupyter

- Implemented a **Stable Diffusion** inference pipeline from scratch in **PyTorch** — VAE encoder/decoder for latent space compression, **U-Net** with multi-head cross-attention for iterative denoising, and **CLIP ViT** text conditioning.
- Mapped pre-trained SD v1.5 weights into the custom architecture and built both **text-to-image** and **image-to-image** generation pipelines.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Rust, Java, C#, Kotlin

Frameworks/Tools: React, Node.js, GCP, Firebase, Android, Docker, Amplitude, PyTorch, TensorFlow

Databases: PostgreSQL, MySQL, Oracle